

Hero Adjusted Plus–Minus in *Marvel Rivals*

Estimates from 317,674 competitive matches

Season 19, PC ranked. Prepared 8 September 2026. Commit 87423d6.

Specification. One observation per match; the outcome is victory for side 0. Each hero enters as a signed contrast (own-side weight less opposing-side weight) under play-time attribution, reduced onto a sum-to-zero basis so every hero receives both a coefficient and a standard error. Controls: a side intercept, the pre-match rank-score differential, composition-shape indicators, and 100 team-up contrasts. Estimated by unpenalised logit with heteroskedasticity-robust (HC1) standard errors.

Why estimates are reported within role. The raw coefficients are dominated by a role-composition effect: role alone explains 64.6% of their variance, and the twelve highest-ranked heroes are all Tanks. This is structural. A team’s tank count is the *sum* of its tank heroes’ indicators, so it is perfectly collinear with the hero block; the composition-shape controls absorb only the categorical part, and the linear part loads onto the individual hero coefficients. The implied direction is also wrong (Table 2). Subtracting each hero’s own-role mean is a linear contrast whose covariance follows exactly by the delta method, and it restores agreement with an assumption-light benchmark: rank correlation with play-time-weighted raw win rates rises from 0.70 to 0.95.

Table 1: Hero adjusted plus–minus, relative to an average hero of the same role

	APM	Std. error	95% CI	W2
<i>Panel A: Tank (15 heroes)</i>				
Peni Parker	7.54***	(0.585)	[6.39, 8.68]	10.06
The Thing	6.92***	(0.289)	[6.35, 7.48]	4.82
The Hood	6.77***	(0.236)	[6.31, 7.23]	0.55
Magneto	3.70***	(0.215)	[3.28, 4.12]	0.62
Hulk	3.50***	(0.342)	[2.83, 4.17]	2.00
Angela	2.92***	(0.399)	[2.14, 3.71]	2.15
Thor	0.85***	(0.323)	[0.22, 1.49]	0.86
Venom	-1.34***	(0.343)	[-2.01, -0.67]	-1.21
Rogue	-1.93***	(0.522)	[-2.95, -0.91]	-3.82
Captain America	-2.23***	(0.577)	[-3.36, -1.10]	-0.65
Devil Dinosaur	-2.66***	(0.362)	[-3.37, -1.95]	-0.82
Groot	-3.81***	(0.467)	[-4.73, -2.90]	-1.77
Deadpool (Vanguard)	-6.26***	(0.374)	[-6.99, -5.52]	-5.42
Emma Frost	-6.95***	(0.266)	[-7.47, -6.43]	-4.75
Doctor Strange	-7.02***	(0.294)	[-7.59, -6.44]	-2.61
<i>Panel B: Damage (27 heroes)</i>				
Magik	14.65***	(0.334)	[14.00, 15.31]	7.97
Black Cat	11.38***	(0.404)	[10.58, 12.17]	7.60
Storm	10.04***	(0.469)	[9.12, 10.96]	6.38
Daredevil	8.22***	(0.355)	[7.52, 8.91]	5.64
Iron Fist	4.56***	(0.338)	[3.89, 5.22]	2.83
Mister Fantastic	4.14***	(0.482)	[3.20, 5.09]	3.12
Human Torch	4.11***	(0.546)	[3.04, 5.18]	1.20
Spider-man	2.78***	(0.268)	[2.26, 3.31]	1.29
Iron Man	2.71***	(0.424)	[1.88, 3.54]	1.81
Black Panther	2.61***	(0.426)	[1.78, 3.45]	0.94
Deadpool (Duelist)	2.23***	(0.339)	[1.57, 2.89]	0.81
Psylocke	1.65***	(0.527)	[0.62, 2.68]	0.13

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Table 1, continued

	APM	Std. error	95% CI	W2
Blade	1.33***	(0.460)	[0.43, 2.23]	0.47
Hela	1.26***	(0.298)	[0.68, 1.84]	1.02
Wolverine	0.63	(0.461)	[-0.27, 1.54]	0.87
Elsa Bloodstone	0.32	(0.347)	[-0.36, 1.00]	-0.16
Winter Soldier	0.09	(0.214)	[-0.33, 0.51]	-0.16
Black Widow	-1.03***	(0.306)	[-1.63, -0.43]	-0.85
Moon Knight	-2.93***	(0.427)	[-3.77, -2.09]	-1.78
Hawkeye	-2.96***	(0.572)	[-4.08, -1.84]	-1.37
Star-lord	-4.13***	(0.447)	[-5.00, -3.25]	-2.58
Scarlet Witch	-5.35***	(0.529)	[-6.39, -4.32]	-4.09
Cyclops	-7.29***	(0.298)	[-7.87, -6.71]	-5.19
Namor	-7.66***	(0.321)	[-8.29, -7.03]	-3.96
The Punisher	-10.69***	(0.374)	[-11.42, -9.95]	-5.51
Phoenix	-13.90***	(0.495)	[-14.87, -12.93]	-7.40
Squirrel Girl	-16.79***	(0.555)	[-17.87, -15.70]	-9.02
<i>Panel C: Support (13 heroes)</i>				
Mantis	13.67***	(0.269)	[13.14, 14.20]	9.93
Ultron	6.27***	(0.456)	[5.38, 7.17]	6.73
Rocket Raccoon	5.18***	(0.212)	[4.76, 5.60]	4.47
Adam Warlock	4.01***	(0.458)	[3.11, 4.91]	4.08
Loki	1.08***	(0.397)	[0.30, 1.86]	1.73
Cloak & Dagger	0.40*	(0.236)	[-0.06, 0.87]	-1.69
Jubilee	0.32	(0.232)	[-0.13, 0.78]	-1.92
Gambit	-0.16	(0.285)	[-0.72, 0.40]	-1.75
Jeff The Land Shark	-4.91***	(0.333)	[-5.57, -4.26]	-3.02
Invisible Woman	-5.43***	(0.219)	[-5.86, -5.00]	-3.43
Deadpool (Strategist)	-6.48***	(0.302)	[-7.07, -5.89]	-5.32
White Fox	-6.83***	(0.294)	[-7.40, -6.25]	-4.95
Luna Snow	-7.13***	(0.240)	[-7.60, -6.66]	-4.87
Matches	317,674			
Player-matches	3,812,088			
Heroes	55			
Log-likelihood	-198,018.6			

Notes. Units are percentage points of win probability evaluated at a balanced match ($\partial p / \partial x = \beta / 4$). Standard errors in parentheses are HC1 and are propagated through the within-role contrast by the delta method. Stars denote Benjamini–Hochberg false-discovery-rate control across the 55-hero family: * $q < 0.10$, ** $q < 0.05$, *** $q < 0.01$; six heroes are not distinguishable from their role average. The final column repeats the estimate under dominant-hero attribution (W2) rather than play-time attribution (W1); that run used 329,572 matches rather than 317,674, because the crawl continued between runs, so the two columns are not estimated on an identical sample. Sample: 321,337 matches crawled, less 1,423 containing draws (`is.win` = 2), 11 with incomplete play time, and 2,229 below a 240-second forfeit floor. Four team-up contrasts were structurally empty and dropped.

Table 2: Raw win rate by number of Tanks fielded

Tanks	Teams	Win rate (%)	vs. two tanks
0	1,990	18.49	-33.81
1	89,534	39.44	-12.86
2	533,702	52.30	—
3	24,456	41.28	-11.02
4	288	34.72	-17.58

Composition from each player’s play-time-dominant hero. The relationship is an inverted U peaking at the 2-2-2 standard that 76% of teams field, so more Tanks is not better. A term linear in tank counts cannot represent this shape; because teams fielding too few Tanks outnumber those fielding too many by roughly four to one, the fitted slope is dragged positive. This is the artifact the within-role normalisation in Table 1 removes.

Table 3: Specification checks

Check	Result	Interpretation
Placebo (permuted heroes)	0.023 vs 0.881	97.4% of signal destroyed
Sum-to-zero normalisation	-1.1×10^{-16}	holds at machine precision
Rank agreement, raw win rates	$\rho = 0.949$	up from 0.703 before correction
Attribution robustness (W1 vs W2)	$\rho = 0.954$	ranking stable; magnitudes are not

The placebo shuffles hero assignment against outcomes and refits; a placebo that failed would invalidate the specification. Rank agreement is Spearman’s ρ against play-time-weighted raw win rates. Under W2 the ranking is preserved but effect sizes roughly halve (mean absolute difference 2.08 points; Squirrel Girl moves from -16.79 to -9.02), so the *ordering* should be treated as the result and magnitudes as an upper bound.

Table 4: Team-up synergies, largest and smallest

Team-up	Premium	Std. error
SQUIRREL MISSILE	12.22	(1.182)
BLADE OF KHONSHU	11.22	(1.080)
FLAWLESS DESIGN	10.58	(0.991)
KUMIHO PALM	7.90	(0.614)
MR. & MRS. X	7.54	(0.626)
BURNING BULLETS	6.22	(1.115)
... 90 team-ups omitted ...		
COMPREHENSIVE DEFENSE	-1.82	(1.050)
GODS OF THUNDER	-1.95	(1.043)
BINDING TIES	-2.86	(0.912)
ALLIED AGENTS	-3.12	(1.611)

The premium a pair earns *beyond* what its two heroes contribute individually, in percentage points, under W2 attribution. Median absolute effect across all 100 team-ups is 2.08 points. Squirrel Missile is the largest while Squirrel Girl is the weakest Damage hero individually, which is the separation of solo value from pair synergy the specification is built to make.

Limitations.

1. *Magnitudes are attribution-dependent.* Ordering is stable across W1 and W2 ($\rho = 0.954$) but effect sizes roughly halve under W2. Play-time attribution does not *condition* on mid-match hero swaps, but it does *weight* by them, and a losing player abandoning a hero reduces that hero’s weight — so it remains exposed to the outcome indirectly.
2. *These are adjusted associations, not causal effects.* Hero choice is not random. Pre-match rank score controls for general player skill but not for skill on the specific hero, and a one-trick is not a random player assigned that hero. Per-player hero playtime would measure this confounder directly; it was costed and declined, so it is bounded rather than eliminated.
3. *Intervals are analytic, not bootstrapped.* Matches are not independent draws — each contains twelve players and so belongs to twelve overlapping player clusters. The specified player-level cluster bootstrap would require roughly 656 hours at this scale under the current implementation, so HC1 intervals are reported instead. They ignore cross-classified dependence and are therefore *too narrow*; point estimates are unaffected.
4. *The sample is rank-selected.* Profile privacy rises from 0% below 4,000 rank score to 70% above 5,000, and the crawl can only expand through public players, so high-elo matches are structurally under-sampled. Further crawling does not fix this.
5. *Four team-ups are absent.* Hero id 1057 is base “Deadpool”, but the roster records only his three role variants, so 1057 is played zero times. The four team-ups defined against it are structurally empty and were dropped rather than mapped onto the variants, which would be an unverifiable modelling assumption.